

YOURSTORY



snowflake

H2S



COCO CLI

HACKATHON

GCC Edition



Team Name : Supply Chain Forge

Team Leader Name : Harsh Patel

Team Size : 1

Problem Statement : Supply Chain Ontology and Governed Conversational Analytics

Problem brief: four systems, four versions of the truth

The business problem

Supply chain data sits in four systems (ERP, WMS, TMS, SRM) that name and date things their own way. Ask “What is our on-time delivery rate?” and each team gets a different answer.

Who it is for

Production planners, procurement leads (buyers) and logistics coordinators, plus the leaders who need one number they can act on.

Pain today → with Supply Chain Forge

Teams reconcile spreadsheets and argue about whose number is right. Now every metric is defined once in a Snowflake semantic view, and people, dashboards and the AI agent all read that one definition.

Industry context

Manufacturing and distribution: 12 plants in APAC, EMEA and AMER, 150 suppliers, 1,200 parts, ~725K orders and ~780K shipments over 10 years, with SAP-style source tables and real-world defects.

Leadership asks: “What is our on-time delivery rate?”

Planning says

68.2%

using the ERP order date

Logistics says

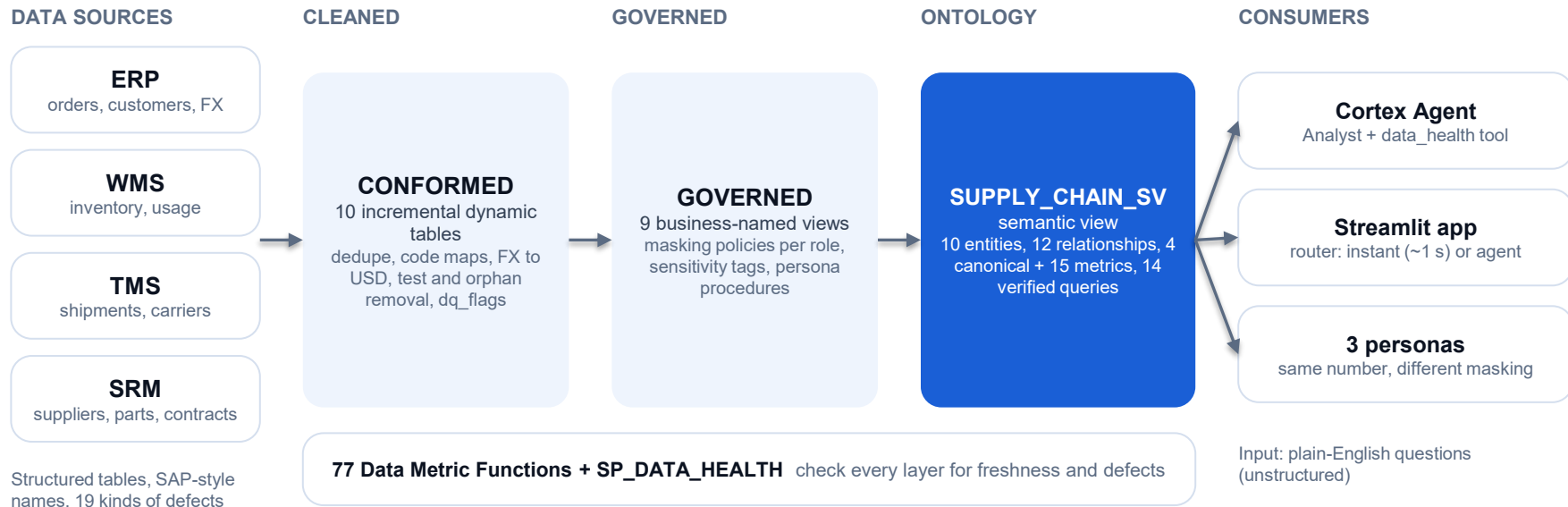
87.5%

using the carrier’s promised date

Same shipments, 19.3 points apart.

Supply Chain Forge makes 87.5% the one governed answer, for every team.

Architecture: one governed layer, built and run in Snowflake



BUILT WITH COCO CLI SKILLS, EACH OWNING ONE LAYER



Impact: one trusted number, answered in seconds

0.000000

difference between Planner, Buyer and Logistics on all 4 metrics (6 decimal places)

19.3 pts

of cross-team disagreement on on-time delivery removed (68.2% vs 87.5%)

~1 s

instant governed answers with no LLM call; agent answers ~12 s at p50

28 / 30

agent evaluation questions correct, incl. multi-part, Hindi and refusals

Measurable outcomes

- One definition per metric replaces per-team SQL and spreadsheet reconciliation
- 93/93 data-quality self-checks pass; 77 DMFs watch every load
- Masking enforced by Snowflake policies, not by the app

Scalability

- Incremental dynamic tables and an XS warehouse handle ~6M source rows
- Rebuilt from the repo into a new account with no hand fix, data byte-identical
- A nightly task appends each new business day automatically

Beyond the demo

- Swap the generator for real SAP and WMS connectors; layers stay the same
- Add a metric or synonym to the semantic view: the agent and app pick it up
- Same pattern fits finance, sales or HR data that disagree across systems

The working prototype: live on Snowflake

Input

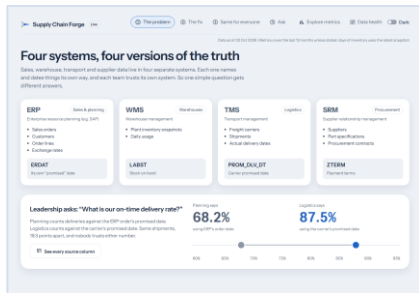
A plain-English question, e.g. "What is on-time delivery by region?"

Processing

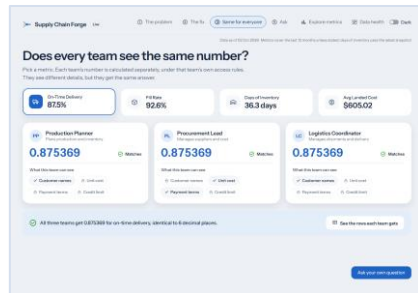
The router runs governed semantic-view SQL or asks the Cortex Agent; masking applies per role

Output

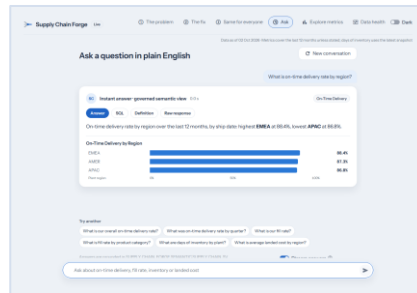
A governed answer with its chart, SQL, metric definition and the path that answered



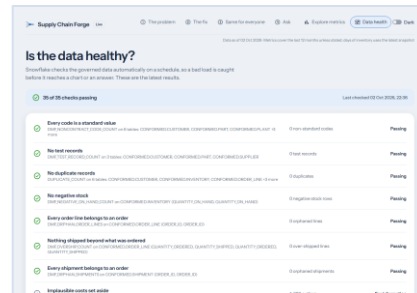
1 The problem: two numbers



2 Same for everyone



3 Ask: instant governed answer



4 Data health: 35/35 passing

Try it: supply-chain-forge.streamlit.app · Code and docs: github.com/Harsh2292/snowflake-forge

Public link on Streamlit Community Cloud, reading Snowflake live through a key-pair service user with a read-only role, rate limits and a Cortex budget.

How it was built: CoCo CLI plus a frozen contract

1

Contract first

docs/CONTRACT.md fixes every object name, metric, valid pairing and masking rule

2

CoCo builds Snowflake

Schemas, CONFORMED dynamic tables, masking, semantic view, Cortex Agent, DMFs, cost caps

3

Handoff lock

Generator, quality and eval SQL written outside Snowflake; CoCo runs it and records each run

4

Proof captured

Real Snowflake output saved as artifacts; ~1,000 tests replay it in CI

What CoCo CLI delivered

- ✓ 4 source schemas, ~6M rows of messy data
- ✓ 10 dynamic tables that clean it incrementally
- ✓ 9 governed views, masking policies for 3 roles
- ✓ 1 semantic view: 19 metrics, 14 verified queries
- ✓ 1 Cortex Agent with a custom data_health tool
- ✓ 77 Data Metric Functions, alerts and cost caps
- ✓ Full replay into a new account with no hand fix

Two AI agents, one frozen contract: CoCo CLI built and ran everything in Snowflake; Claude Code built the app, tests and docs; a human coordinated both.

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THANK YOU

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